

How Nearly Repeating Parity Forces a Return: A Friendly Guide to Finite Collatz Defects

Companion to “Finite Parity Defects Force Collatz Returns”*

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1 The result in ordinary language

Use the shortcut Collatz rule: halve an even number; triple an odd number, add one, and halve it. Record a 1 at each odd number and a 0 at each even number. Now compare this parity diary with a copy shifted forward by q places.

If the two copies disagree only a few times over a sufficiently long finite stretch, the orbit must actually return to an earlier number. The theorem gives both the required stretch length and a short list of checkpoints where the return must occur.

Let K satisfy $N + 1 \leq 2^K$ for the starting number N . For instance, $N = 3$ allows $K = 2$. If there are at most D disagreements, the sufficient number of comparisons is

$$L = (2^{D+1} - 1)(q + K).$$

Those comparisons use $L + q$ parity letters. The possible return checkpoints are

$$s_k = (2^k - 1)(q + K), \quad k = 0, 1, \dots, D.$$

What Lean proves. At one of these checkpoints, the orbit value equals the value q steps later. For $q > 0$, the future repeats and the orbit is bounded. For $q = 2$ and a positive starting number, the theorem goes further: the orbit reaches one.

The result allows disagreements anywhere in the checked interval. It does not assume a particular Sturmian slope or a favorable density of even steps. The cost is that the sufficient length grows exponentially with the allowed number of disagreements.

This is a new proved criterion in the repository. We have not proved that every positive starting number satisfies it.

*Written with Codex (OpenAI). The technical paper and Lean source give the exact statements and proofs. No claim of literature priority is made.

2 Why an exact interval matters

Two distinct starting numbers sharing their first r parity letters differ by a nonzero multiple of 2^r . Their difference is therefore at least 2^r . This is the arithmetic fact behind the earlier exact repetition criterion.

There is also a simple size bound. After u shortcut steps,

$$T^u(N) + 1 \leq 2^u(N + 1) \leq 2^{u+K}.$$

It is deliberately generous, but holds for every choice of parities.

At a time s , compare the two orbit values $T^s(N)$ and $T^{s+q}(N)$. If their parity diaries agree for the next $s + q + K$ letters, the divisibility lower bound is too large for these two values to be distinct. They must be equal.

The local choice. At every checkpoint s , either the orbit returns after q steps, or there is a parity disagreement before time $2s + q + K$.

Set $A = q + K$. Choose checkpoints $0, A, 3A, 7A, \dots$ so that the search windows fit together without overlapping:

Checkpoint	Window in which a defect is forced	Next checkpoint
0	$[0, A)$	A
A	$[A, 3A)$	$3A$
$3A$	$[3A, 7A)$	$7A$
$7A$	$[7A, 15A)$	$15A$

If every one of $D + 1$ checkpoints failed to return, these disjoint windows would contain $D + 1$ different disagreements. That contradicts the allowed total of D .

This replaces a direct attempt to bound swap costs. Moving one parity letter across a late adjacent position can have a very large weighted arithmetic cost. The window argument instead uses the exact stretches between disagreements.

3 Allowing edits to a repeating template

Suppose a finite parity diary differs in at most e letters from a repeating template with period q . Each edited letter can affect at most two comparisons with the shifted copy: one in which it is the first letter, and one in which it is the second.

Thus at most e edited letters cause at most $2e$ shift disagreements. The sufficient comparison length becomes

$$L = (2^{2e+1} - 1)(q + K),$$

and the template is checked for $L + q$ letters. The edit positions can be arbitrary. Lean checks this extension too.

For a concrete example, start at 3:

$$3, 5, 8, 4, 2, 1, 2, 1, \dots$$

Its diary begins 11000101... Compare it with its shift by two places. The only disagreements are at times 0, 1, 3. With $D = 3$, $q = 2$, and $K = 2$, the sufficient comparison length is 60. The checkpoints are 0, 4, 12, 28. Already at time 4, the value 2 equals the value two steps later.

The same diary differs from the template 010101... only at times 0 and 3. This gives $e = 2$ edited letters, whose more conservative template bound is $L = 124$. The distinction matters: edited letters and shift disagreements are different counts.

4 Shorter windows from a sharper height bound

The first theorem allowed one bit of height growth per step. We can improve this without assuming anything new about the orbit. Keeping the added-one correction, the shortcut map satisfies

$$2^u(T^u(N) + 1) \leq 3^u(N + 1).$$

Five steps therefore need at most three bits of growth, since $3^5 = 243 < 256 = 2^8$. Checking the remaining zero to four steps gives

$$T^u(N) < 2^{\lceil 3u/5 \rceil + K}.$$

Use that smaller allowance to choose the checkpoints:

$$\sigma_0 = 0, \quad \sigma_{k+1} = \sigma_k + \left\lceil \frac{3(\sigma_k + q)}{5} \right\rceil + K.$$

The same counting proof works with the shorter horizon $L \geq \sigma_{D+1}$. Edited templates use σ_{2e+1} .

A checked improvement. Lean proves that these horizons are never larger than the original ones. Their geometric growth has an upper bound with factor $8/5 = 1.6$, improving the old factor of 2. The hypotheses about the seed and the allowed defects are unchanged.

For lag two, the worked examples become

Start N	Bit bound K	Defects D	Old horizon	New horizon
3	2	3	60	36
7	3	5	315	128
9	4	6	762	250

Lean checked the new horizons and finite defect counts. These illustrate the theorem; they are not new computational Collatz records.

5 What remains between this and Collatz

The missing step. We know what happens if a finite parity prefix is sufficiently close to periodic. We do not know that every hypothetical counterexample must have such a prefix.

For a general positive period, the return could belong to a nontrivial cycle. For period two, Lean proves that the only positive return values are 1 and 2, so the certificate implies convergence to one. But its universal availability is still unproved. Every orbit reaching one eventually alternates; assuming that all orbits have the needed period-two certificate would simply rephrase the convergence problem.

The shorter windows still require an independent restriction on the number of defects. No such universal restriction has been established. The expanded audit checks 44 selected declarations using only Lean’s standard foundational axioms; it is not an independent kernel replay.

The technical paper is `paper/defects.pdf`, with source `paper/defects.tex`. The formal development is `lean/Collatz/ParityDefects.lean` and `lean/Collatz/DefectHeight.lean`; the research note is `analysis/PARITY_DEFECTS.md`.